When a lower forecast loss does not make a better delivery promise

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1. Question and summary

I started this project wanting to predict which Olist orders would arrive late. Along the way I realised that "late" compares two things: how long delivery takes, and how generous the promised date is. Thus a model can reduce lateness without making delivery faster. The question became: **when does a better forecast actually give a better promised date?**

The data are the public Olist e-commerce orders (Brazil, 2017–2018). I study single-seller orders that were eventually delivered, so every result is conditional on delivery. A promised date is a whole number of days d; an order is covered if it arrives within d days. The target coverage is 90%. The evaluation set E has 19,950 orders purchased from April to July 2018.

I compared a per-state quantile rule, a linear quantile regression and a shallow gradient-boosted model (HGB). All thresholds, data splits and candidate models were written down and hashed before any new model was fitted. Three findings:

1. **The comparison design changed the verdict.** In the project's earlier comparison, regression promised 2.57 more days than the state rule. With the same training data, the same calibration procedure and the same feature timing, it promised 1.54 fewer days at 0.31 points lower coverage.
2. **A lower forecast loss did not give shorter dates.** HGB had a 6.5% lower quantile loss than the linear model, yet its calibrated dates were 1.22 days longer.
3. **No new policy met the pre-set bar for replacing the existing rule.** All calibrated policies over-covered E (96–97% against a 90% target).

These results come from one evaluation period that the project had already analysed. Therefore this is a pre-registered exploratory study, not a blind test.

2. Loss and decision: why a quantile, not the mean

If a late day and a spare day cost different amounts, the best promise is a quantile of the delivery time, not its mean. Let Y be the delivery time in whole days and d the promise. Consider the simple loss

$$L(d, Y) = c_{late} \cdot (Y - d)_+ + c_{slack} \cdot (d - Y)_+.$$

The first term charges each late day; the second charges each unused day of slack. Let $R(d) = E[L(d, Y)]$ and F the distribution function of Y. Moving the promise from d to $d + 1$ saves c_{late} on every order with $Y > d$ and adds c_{slack} on every order with $Y \leqslant d$. Therefore

$$R(d + 1) - R(d) = (c_{late} + c_{slack}) \cdot F(d) - c_{late}.$$

This difference never decreases in d. Thus R is minimised at the smallest whole d with $F(d) \geqslant q$, where $q = c_{late} / (c_{late} + c_{slack})$. That is exactly the empirical quantile rule the project uses (NumPy's inverted_cdf). The pinball loss at level q is this loss with $c_{late} = q$ and $c_{slack} = 1 - q$ (scikit-learn). So a 90% target corresponds to a 9 : 1 penalty ratio. By contrast, the mean minimises squared error and the median minimises a 1 : 1 absolute loss; both treat a late day and a spare day alike.

A check on validation data. On the 12,045 validation orders (July–September 2017), I gave every order the same promise and computed the 9 : 1 loss for $d = 1, \ldots, 60$. The minimum is at $d = 20$, which is the empirical 90% quantile: $F(19) = 89.6\% < 90\% \leqslant F(20) = 91.4\%$.

Promise	d (days)	On time	9 : 1 loss per order
Mean (11.43), rounded up	12	65.5%	23.76
Median	10	53.1%	30.51
Empirical 90% quantile	20	91.4%	15.61

What 9 : 1 does not mean. It follows from choosing $q = 0.90$; I did not measure it. It is not Olist's real cost ratio, and I have no cost data. Customer harm may not grow linearly with late days; if one late day already does most of the damage, a coverage constraint is closer to the truth than a linear cost. Costs may also differ by customer and product. Therefore the study reports the trade-off between coverage and promised days. It does not claim a profit-optimal date.

3. Comparison design

A fair comparison changes only the method. In the fair comparison, each calibrated policy therefore used the same training orders (F), the same label cut-off, the same rule for seller-history features and the same calibration procedure on C. The uncalibrated controls (B1 and the retained rule B0) are kept apart. A label counts only if the order was delivered before the decision date.

Segment	Purchases	Labels known before	Orders used	Late rate*	Role
F0	Jan–Jun 2017	2017-12-01	13,616	3.89%	fit candidates
V	Jul–Sep 2017	2017-12-01	12,045	3.38%	select settings
F	Jan–Sep 2017	2018-04-01	25,666	3.67%	final fit
C	Oct–Dec 2017	2018-04-01	17,056	8.74%	calibration shift
E	Apr–Jul 2018	—	19,950	4.26%	final scoring run

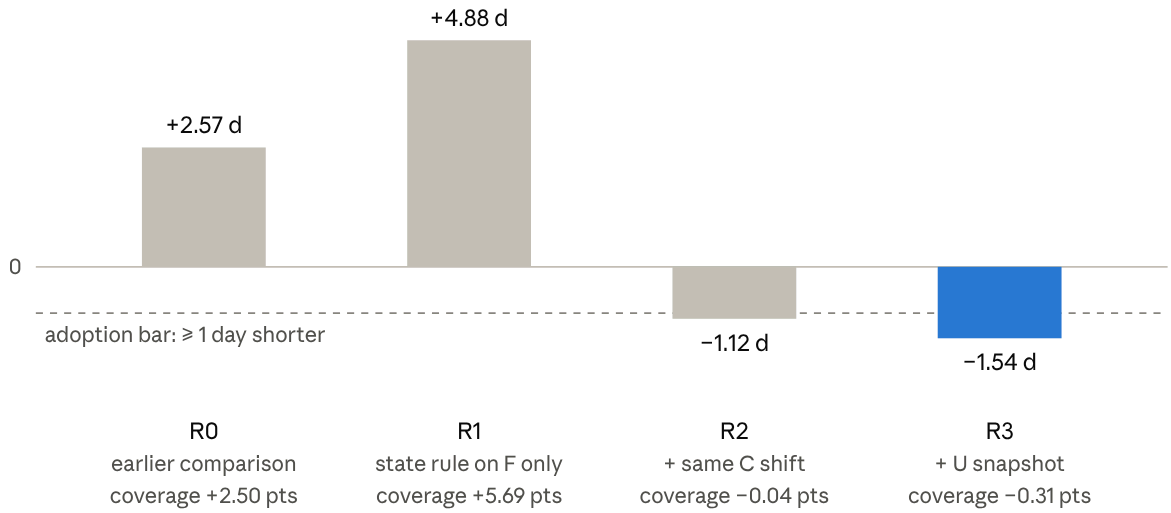
*Against Olist's own estimated date.

Calibration. Each calibrated policy is shifted by its own number computed on C: the k-th smallest of its residuals $y - \hat{y}$, with $k = \lceil (n + 1) \cdot q \rceil$ and $n = 17{,}056$. The promise is $\max\{1, \lceil \hat{y} + \text{shift} \rceil\}$. This has the form of split conformal, but its coverage guarantee needs C and E to be exchangeable. C had a much higher late rate than E, so I do not claim a guarantee (Barber et al.).

Stepwise comparison. The project's earlier comparison (R0) differed in three conditions at once. So I changed one condition per step and recorded the gap "regression minus state rule" each time. The pre-set primary comparison is R3 against R0; R1 and R2 are descriptive.

Making the comparison fair flipped the gap from +2.57 to −1.54 days

Regression minus state rule, mean promised days on E (19,950 orders); one condition changed per step



outputs/e1/score/h1_steps.csv · E, April–July 2018, 19,950 orders

Under this fixed order, changing the design alone reversed the judgement. The largest step is R2, where the state rule receives the same C shift as the regression (+6 days for the state rule). This is a design-sensitivity result, not a causal statement about which method is better.

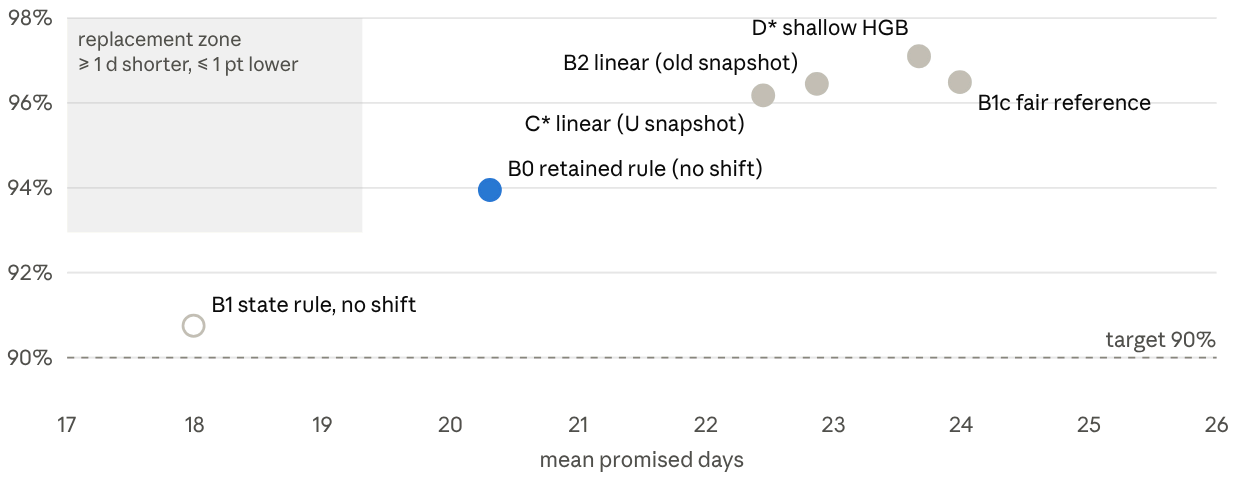
Two references are kept apart. The **fair reference** (B1c: state rule fitted on F with the same C shift) asks whether a model adds value under equal conditions. The **replacement reference** (the project's retained state rule B0, fitted on F + C without a shift) asks whether the existing rule is worth replacing.

4. Results

The E1 models and decision rules were frozen before the final E1 scoring run, and each policy was scored once in that run; E had already been examined in Stage 7. A policy replaces another only if its mean promise is at least 1 day shorter and its coverage at most 1 point lower, with no month of ⩾ 1,000 orders more than 3 points below target. Intervals are 95% week-block bootstrap intervals (17 purchase weeks, 2,000 draws).

No calibrated policy lands in the zone that would replace the retained rule B0

E, 19,950 orders, q = 0.90. Filled grey: same C shift step; hollow: no shift; blue: retained rule B0



outputs/e1/score/policy_table.csv · E, 19,950 orders · fixed, pre-registered policies only (no post-hoc sweep)

Question	Comparison	Result	Pre-set verdict
H1: does the fair design change the judgement?	R3 vs R0	gap moved by −4.11 days	met
H2: does a linear change help?	C* vs B3	none selected on V (C* = C0 = B3); g = 0 by construction	not supported
H3: does shallow HGB help?	D* vs B1c	−0.32 days (−0.55 to −0.17), coverage +0.61 pts	below 1-day bar
Replace the retained rule?	C*, D* vs B0	+2.15 and +3.37 days	no

A separate check at q = 0.9126. This target matches Olist's coverage on C, so it compares with Olist's own promise; it is reported apart from the q = 0.90 results. C* promised 23.42 days at 96.62% coverage, against Olist's 24.44 days at 95.74%. It therefore passes the pre-set rule, but only by 0.02 days; D* (24.75 days) and B1c (25.85 days) fail. A post-hoc interval for C*'s day difference is −3.25 to +0.69 days, which includes zero. So I read this as weak evidence, not as a confirmed advantage over Olist.

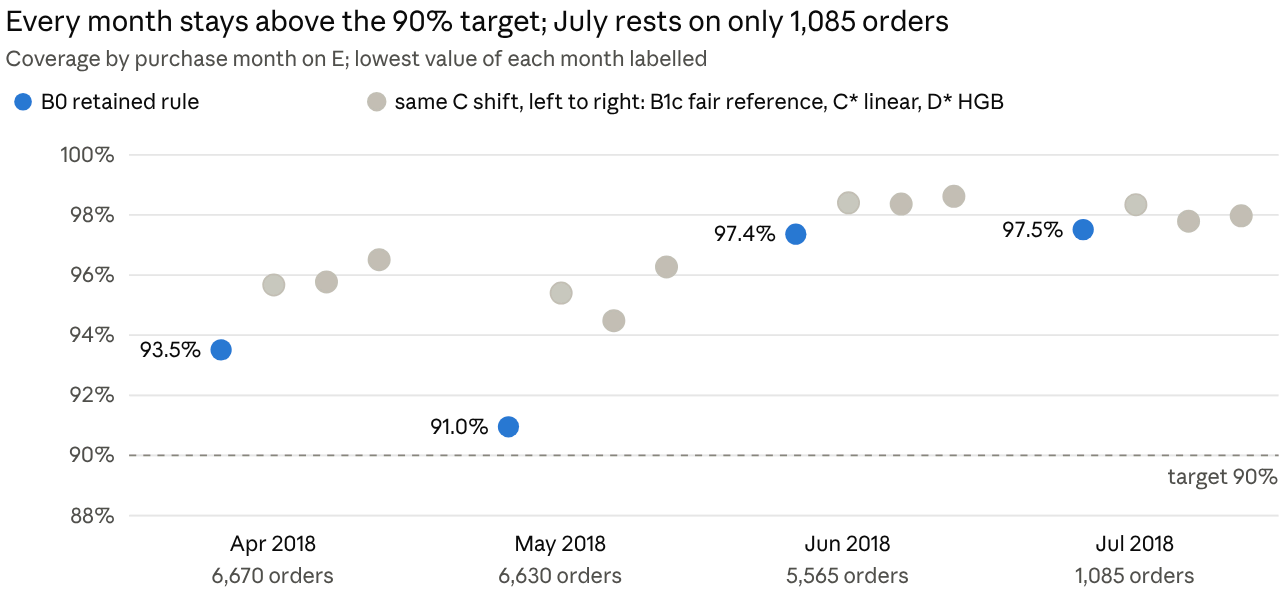
5. Interpretation and limits

Why the HGB dates are longer. HGB's uncalibrated integer-day pinball loss was about 6.5% lower than the linear model's (1.324 against 1.417). However, its unrounded predictions were already about 0.90 days longer on average. The calibration shift then added about 0.31 days to the gap, and rounding added about 0.006 days.

Days	Linear C*	HGB D*	D* − C*
Mean unrounded prediction	15.809	16.709	+0.900
Shift computed on C	6.148	6.459	+0.311
Rounding and the 1-day floor	0.497	0.503	+0.006
Mean promised days	22.453	23.671	+1.217

Each value is rounded from full precision, so the rows may not add exactly. Thus lowering the quantile loss and shortening the average promise are different goals. Before calibration, the linear dates covered 88.33% of orders and the HGB dates 91.08%. Because the loss charges a late day nine times a spare day, longer dates with fewer misses can score better. This is not a reason to pick the uncalibrated HGB after the fact.

Over-coverage. The policies that share the C shift step covered 96–97% against a 90% target. The uncalibrated controls (B1 and the retained B0) are outside this statement.



outputs/e1/score/monthly.csv · E, April–July 2018

This is consistent with calibration scores that changed over time. However, the late rate against Olist's own date fell from 8.74% on C to 4.26% on E, and that rate depends on delivery times, on how generous the original dates were, and on the mix of orders. This study has not separated the three. B0 also differs from the new policies in how it uses the training data (F + C, no shift). Therefore the conclusion is "do not replace in this round", not "the models or the calibration do not work". Calibration is the part most worth studying next, but I have not shown it is the main cause.

Other limits.

- One evaluation period, already analysed in stage 7: a pre-registered exploratory study, not a blind test.
- Conditional population: single-seller orders that were eventually delivered.
- The week-block intervals use 17 weeks and describe fixed models only; they exclude refitting, tuning and selection uncertainty.
- The search over models was a small, stepwise one. It can miss changes that only help together.
- Promise dates do not change logistics. Nothing here shows faster delivery or happier customers.

6. Conclusion

This round does not support replacing the retained state rule. The more useful results are about method. First, a fair comparison rule changed the verdict between two simple methods. Second, a 6.5% lower forecast loss did not turn into shorter promises. Third, I accepted that the more complex model did not meet the bar I set before seeing the results. The open question is how well a calibration shift learned in one period carries over to the next. E1 ends here; that question would need a new, separately registered study.

Supplementary material, a reproducibility appendix and a contribution statement are in the companion document “Supplement”, sent as a separate file (the second tab of the online version).